

Notes for Project

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1 7/02

let

$$\dot{\theta}_i = \omega - \operatorname{sgn}\left(\frac{\partial V(\theta)}{\partial \theta_i}\right) = \omega - \operatorname{sgn}\left(\sum_{j=1}^N \sin(\theta_i - \theta_j)\right)$$

subject to $\theta_i(0) = \theta_i^0$

let

$$\dot{\theta} = X(\theta)$$

Then $X : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a piece-wise continuous vector field on $|\theta_i - \theta_j| = n\pi, n \in \mathbb{N}$ **(?A**

We assume the initial condition of $\theta^0 = (\theta_1^0, \dots, \theta_N^0)$ has the property that

$$D(\theta^0) = \max(\theta^0) - \min(\theta^0) < \pi$$

Our goal is to show that under the constraint the phase-frequency synchronization occurs in $D(\theta^0)/2$

We can consider the simple case that $N=2$. The system thus can be reduced as

$$X(\theta) = \begin{bmatrix} \omega - \operatorname{sgn}(\sin(\theta_1 - \theta_2)) \\ \omega - \operatorname{sgn}(\sin(\theta_2 - \theta_1)) \end{bmatrix} = \begin{cases} \begin{bmatrix} \omega - 1 \\ \omega + 1 \end{bmatrix}, & \text{if } \theta_1 > \theta_2 \\ \begin{bmatrix} \omega + 1 \\ \omega - 1 \end{bmatrix}, & \text{if } \theta_2 > \theta_1 \end{cases}$$

Define

$$F[X](x) = \bigcap_{\delta > 0} \bigcap_{\mu(S)=0} \operatorname{co}\{X(B(x, \delta) - S)\}$$

Definition 1. An absolute continuous function $\phi : [a, b] \rightarrow \Omega$ is a Filippov solution to $\dot{x} = f(x)$ if $\dot{\phi}(t) \in F[f](\phi(t))$ for almost all $t \in [a, b]$

To begin with, we should check its existence of filippov solution

Lemma 1. *For any system described above, there is a filippov solution.*

Proof. see theorem 8 [pg. 85] □

Lemma 2. *Let $\theta(t)$ be any Filippov solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$. If there exists a time T such that $\theta(T) \in \Sigma$, then $\theta(\tau) \in \Sigma, \forall \tau \geq T$ where $\Sigma = \{\theta = (\theta_1, \theta_2) | \theta_1 = \theta_2\}$*

Proof. Let a function $H(\theta) = \theta_1 - \theta_2$. Note that $\Sigma = \{\theta | H(\theta) = 0\}$

We need to check the derivative of $H(\theta)$ along the trajectory, i.e. $\frac{d}{dt}H(\theta(t))$ at $\theta_1 = \theta_2$. Since $H(\theta)$ is continuously differentiable, we have $\frac{d}{dt}H(\theta(t)) = \nabla H(\theta) \cdot \dot{\theta}(t)$ with $\dot{\theta}(t) \in F[X](\theta(t))$ almost for all t . Thus,

$$\frac{d}{dt}H(\theta(t)) = [1, -1] \cdot \dot{\theta}(t)$$

$$\text{At } \theta(T) = (\theta_1(T), \theta_2(T)), \quad \dot{\theta}(T) = \begin{bmatrix} \omega \\ \omega \end{bmatrix} \cdot \frac{d}{dt}H(\theta(T)) = [1, -1] \begin{bmatrix} \omega \\ \omega \end{bmatrix}$$

$$\frac{d}{dt}H(\theta(T)) = 0$$

This means $\int_T^t \frac{d}{ds}H(\theta(s))ds = H(\theta(T)) - H(\theta(t)) = 0$ with $\theta(T) \in \Sigma$

$$H(\theta(t)) = H(\theta(T)), \quad \forall t \in [T, T + \delta]$$

We also need to show $\delta = +\infty$. We assume the contrary. There exists a time $\tilde{t}$ such that $H(\theta(T)) \neq H(\theta(\tilde{t}))$ and $H(\theta(t)) = H(\theta(T)), \forall t \in [T, \tilde{t})$. Since H, θ are continuous, $\lim_{t \rightarrow \tilde{t}} H(\theta(t)) = H(\theta(\tilde{t})) = H(\theta(T))$

Hence, $H(\theta(T)) = H(\theta(t)), \forall t \in [T, +\infty)$ □

Lemma 3. *Let $\theta(t)$ be any solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$ and $\theta^0 \in D_1 = \{\theta | \theta_1 < \theta_2\}$ (or $D_2 = \{\theta | \theta_2 < \theta_1\}$). The solution will not cross Σ , i.e. the solution will stay at $\overline{D_1}$ (or $\overline{D_2}$)*

Proof. Since $\theta(t)$ continuous, suppose at some time $\tilde{t}$ with $\theta(\tilde{t}) \in D_2$. By IVT, there is a time $\tau \in [0, \tilde{t}]$ such that $\theta(\tau) \in \Sigma$. By lemma 2, the solution afterwards will stay at Σ , which is a contradiction. □

Theorem 1. *Let $\theta(t)$ be any solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$ and $\theta^0 \in D_1 = \{\theta | \theta_1 < \theta_2\}$ (or $D_2 = \{\theta | \theta_2 < \theta_1\}$). Then, $\theta(t)$ will reach Σ exactly at $t = D(\theta^0)/2$*

Proof. Let $\theta(t)$ reaches Σ at $t = \alpha$. Suppose the solution does not reach Σ at $D(\theta^0)/2$ but at some time $\alpha > D(\theta^0)/2$. The solution will lie in D_1 which is a continuous vector field, i.e.

$$\dot{\theta} = X(\theta) = \begin{bmatrix} \omega + 1 \\ \omega - 1 \end{bmatrix}, \quad \theta(0) = \theta^0, \quad \forall t \in [0, \frac{D(\theta)}{2}]$$

Solving the ode, we get

$$\int_0^{\frac{D(\theta^0)}{2}} \dot{\theta}(t) dt = \int_0^{\frac{D(\theta^0)}{2}} X(\theta(s)) ds$$

$$\theta\left(\frac{D(\theta^0)}{2}\right) = \theta^0 + \left[\frac{\omega + 1}{\omega - 1} \right] \frac{D(\theta)}{2} = \left[\frac{\theta_1^0 + \frac{(\omega+1)(\theta_2^0 - \theta_1^0)}{2}}{\theta_2^0 + \frac{(\omega-1)(\theta_2^0 - \theta_1^0)}{2}} \right] = \left[\frac{(\omega+1)\theta_2^0 - (\omega-1)\theta_1^0}{(\omega+1)\theta_2^0 - (\omega-1)\theta_1^0} \right]$$

As we can observed, $\theta_1(\frac{D(\theta)}{2}) = \theta_2(\frac{D(\theta)}{2})$ which is a contradiction.

Then suppose the opposite, $\alpha < D(\theta^0)/2$. Since $\theta(t)$ is continuous, we have $\lim_{x \rightarrow \alpha} \theta(x) = \theta(\alpha)$. That is,

$$\theta(x) = \theta^0 + \left[\frac{\omega + 1}{\omega - 1} \right] x = \theta(\alpha), \text{ as } x \rightarrow \alpha$$

Since $\theta_1(\alpha) = \theta_2(\alpha)$, $\theta_1^0 + (\omega + 1)x = \theta_2^0 + (\omega - 1)x$. Solving for x , we get $x = (\theta_2^0 - \theta_1^0)/2 = D(\theta^0)/2$, which is a contradiction.

Thus, we get $\alpha = D(\theta)/2$ □

2 7/09

Consider $N=n$ oscillators situation.

$$X(\theta) = \begin{bmatrix} \omega - \text{sgn}(\sum_{k=1}^n \sin(\theta_1 - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^n \sin(\theta_n - \theta_k)) \end{bmatrix}$$

$$\begin{aligned} \text{Note that } \sum_{k=1}^n \sin(\theta_i - \theta_k) = 0 &\implies \sum_{k=1}^n (\sin(\theta_i) \cos(\theta_k) - \cos(\theta_i) \sin(\theta_k)) = 0 \\ \implies \sum_{k=1}^n (\sin(\theta_i) \cos(\theta_k)) &= \sum_{k=1}^n (\cos(\theta_i) \sin(\theta_k)) \implies \tan(\theta_i) = \frac{\sum_{k=1}^n \sin(\theta_k)}{\sum_{k=1}^n \cos(\theta_k)} \end{aligned}$$

Split the space to some regions with discontinuous plane, which are $\Sigma_i = \{\theta \mid \tan(\theta_i) = \frac{\sin(\theta_1) + \sin(\theta_2) + \dots + \sin(\theta_n)}{\cos(\theta_1) + \cos(\theta_2) + \dots + \cos(\theta_n)}\}$

Lemma 4. *For any system described above, there is a filippov solution.*

Proof. see theorem 8 [pg. 85] □

Lemma 5. *Let $\theta(t)$ be any Filippov solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$. If there exists a time T such that $\theta(T) \in \Sigma_i$, then $\theta(\tau) \in \Sigma_i, \forall \tau \geq T$*

Proof. □

3 7/23

Consider n=3 case

$$X(\theta) = \begin{bmatrix} \omega - \operatorname{sgn}(\sin(\theta_1 - \theta_2) + \sin(\theta_1 - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_2 - \theta_1) + \sin(\theta_2 - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_3 - \theta_1) + \sin(\theta_3 - \theta_2)) \end{bmatrix}$$

We note that $\Sigma = \{\theta | (\theta_1 - \frac{\theta_2}{2} - \frac{\theta_3}{2})(\theta_2 - \frac{\theta_1}{2} - \frac{\theta_3}{2})(\theta_3 - \frac{\theta_1}{2} - \frac{\theta_2}{2}) = 0\}$ is discontinuous, which splits the space into 6 areas.

-	-	-	IMPOSSIBLE case
+	-	-	$X(\theta) = [\omega - 1, \omega + 1, \omega + 1]^T$
-	+	-	$X(\theta) = [\omega + 1, \omega - 1, \omega + 1]^T$
-	-	+	$X(\theta) = [\omega + 1, \omega + 1, \omega - 1]^T$
-	+	+	$X(\theta) = [\omega + 1, \omega - 1, \omega - 1]^T$
+	-	+	$X(\theta) = [\omega - 1, \omega + 1, \omega - 1]^T$
+	+	-	$X(\theta) = [\omega - 1, \omega - 1, \omega + 1]^T$
+	+	+	IMPOSSIBLE case

Lemma 6. Let $\theta(t)$ be any Filippov solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$. If there exists a time T such that $\theta(T) \in \Sigma$, then $\theta(\tau) \in \Sigma, \forall \tau \geq T$

Proof. Let $H(\theta) = (x - \frac{y}{2} - \frac{z}{2})(y - \frac{x}{2} - \frac{z}{2})(z - \frac{x}{2} - \frac{y}{2})$. Assume at some point τ such that $H(\tau) \in \Sigma$

We need to check $\frac{d}{dt}H(\tau) = \nabla H(\theta) \cdot \dot{\theta}(\tau) = \dots$

Wlog, assume $\theta_1 = \theta_2 = \theta_*$

$$\frac{d}{dt}H(\tau) = \left[\frac{-3}{8}(\theta_2 - \theta_3)^2, \frac{-3}{8}(\theta_2 - \theta_3)^2, \frac{3}{4}(\theta_2 - \theta_3)^2 \right] \begin{bmatrix} \omega - \operatorname{sgn}(\sin(\theta_* - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_* - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_3 - \theta_*) + \sin(\theta_3 - \theta_*)) \end{bmatrix}$$

□

let $\Sigma_* = \{\theta | (\theta_1 - \theta_2)(\theta_1 - \theta_3)(\theta_2 - \theta_3) = 0\}$

Lemma 7. Let $\theta(t)$ be any Filippov solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$. If there exists a time T such that $\theta(T) \in \Sigma_*$, then $\theta(\tau) \in \Sigma_*, \forall \tau \geq T$

Proof. Let $H(\theta) = (\theta_1 - \theta_2)(\theta_1 - \theta_3)(\theta_2 - \theta_3)$. Assume there exists a τ such that $\theta(\tau) \in \Sigma_*$, i.e. $H(\theta(\tau)) = 0$ or $\theta_i = \theta_j$ for $i, j = 1, 2, 3$

We need to check $\frac{d}{dt}H(\theta(\tau)) = \nabla H(\theta) \cdot \dot{\theta}(\tau) =$

$$[(2\theta_1 - \theta_2 - \theta_3)(\theta_2 - \theta_3), (\theta_1 - \theta_3)(\theta_1 - 2\theta_2 + \theta_3), -(\theta_1 - \theta_2)(\theta_1 + \theta_2 - 2\theta_3)] \begin{bmatrix} \omega - \operatorname{sgn}(\sin(\theta_1 - \theta_2) + \sin(\theta_1 - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_2 - \theta_1) + \sin(\theta_2 - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_3 - \theta_1) + \sin(\theta_3 - \theta_2)) \end{bmatrix}$$

Wlog, assume $\theta_1 = \theta_2 = \theta_*$.

$$\frac{d}{dt}H(\theta(\tau)) = [(\theta_2 - \theta_3)^2, -(\theta_2 - \theta_3)^2, 0] \begin{bmatrix} \omega - \operatorname{sgn}(\sin(\theta_* - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_* - \theta_3)) \\ \omega - \operatorname{sgn}(\sin(\theta_3 - \theta_*) + \sin(\theta_3 - \theta_*)) \end{bmatrix} = 0$$

This means $\int_T^t \frac{d}{ds}H(\theta(s))ds = H(\theta(T)) - H(\theta(t)) = 0$ with $\theta(T) \in \Sigma$

$$H(\theta(t)) = H(\theta(T)), \forall t \in [T, T + \delta]$$

We also need to show $\delta = +\infty$. We assume the contrary. There exists a time $\tilde{t}$ such that $H(\theta(T)) \neq H(\theta(\tilde{t}))$ and $H(\theta(t)) = H(\theta(T))$, $\forall t \in [T, \tilde{t}]$. Since H, θ are continuous, $\lim_{t \rightarrow \tilde{t}} H(\theta(t)) = H(\theta(\tilde{t})) = H(\theta(T))$

Hence, $H(\theta(T)) = H(\theta(t))$, $\forall t \in [T, +\infty)$ $\square$

4 7/26

In $n=3$, we can split the space into 6 regions, which are

$$\begin{aligned} D_1 &= \{\theta | \theta_1 > \theta_2 > \theta_3\} \\ D_2 &= \{\theta | \theta_1 > \theta_3 > \theta_2\} \\ D_3 &= \{\theta | \theta_2 > \theta_1 > \theta_3\} \\ D_4 &= \{\theta | \theta_2 > \theta_3 > \theta_1\} \\ D_5 &= \{\theta | \theta_3 > \theta_1 > \theta_2\} \\ D_6 &= \{\theta | \theta_3 > \theta_2 > \theta_1\} \end{aligned}$$

By lemma 7, we know that solutions will not cross Σ_*

Lemma 8. *Let $\theta(t)$ be any Filippov solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$ and $\theta^0 \in D_i$. The solution will stay at $\overline{D_i}$, $\forall i = 1, 2, \dots, 6$*

Proof. Since there exists a continuous solution of the system, suppose $\theta^0 \in D_i$ and suppose at some time $\tilde{t}$ with $\theta(\tilde{t}) \in D_j$ ($j \neq i$). By IVT, there is a time $\tau \in [0, \tilde{t}]$ such that $\theta(\tau) \in \Sigma_*$. By lemma 7, the solution will stay at Σ_* , which is a contradiction. $\square$

Theorem 2. *Let $\theta(t)$ be any solution of system with initial phase diameter satisfying $D(\theta^0) < \pi$ and $\theta^0 \in D_i$. Then, $\theta(t)$ will reach $\theta_1 = \theta_2 = \theta_3$ exactly at $t = D(\theta^0)/2$*

Proof. Let $\theta(t)$ reaches $\theta_1 = \theta_2 = \theta_3$ at $t = \alpha$. Suppose the solution does not reach $\theta_1 = \theta_2 = \theta_3$ at $D(\theta^0)/2$ but at some time $\alpha > D(\theta^0)/2$. The solution will lie in $\overline{D_i}$, $\forall i = 1, 2, \dots, 6$. Wlog, we suppose $\theta^0 \in D_1$. We have

$$\dot{\theta}(t) = \begin{bmatrix} \omega - 1 \\ \omega - \operatorname{sgn}(\sin(\theta_2 - \theta_1) + \sin(\theta_2 - \theta_3)) \\ \omega + 1 \end{bmatrix}$$

Solving the ode, we get,

$$\int_0^{\frac{D(\theta^0)}{2}} \dot{\theta}(t) dt$$

$$\theta\left(\frac{D(\theta^0)}{2}\right) = \theta^0 + \begin{bmatrix} \omega + 1 \\ k(t) \\ \omega - 1 \end{bmatrix} \frac{D(\theta)}{2} = \begin{bmatrix} \theta_1^0 + \frac{(\omega+1)(\theta_2^0 - \theta_1^0)}{2} \\ k'(t) \\ \theta_2^0 + \frac{(\omega-1)(\theta_2^0 - \theta_1^0)}{2} \end{bmatrix} = \begin{bmatrix} \frac{(\omega+1)\theta_2^0 - (\omega-1)\theta_1^0}{2} \\ k'(t) \\ \frac{(\omega+1)\theta_2^0 - (\omega-1)\theta_1^0}{2} \end{bmatrix}$$

From lemma 8, we know that $\theta_1(t) \geq \theta_2(t) \geq \theta_3(t) \forall t$. Since at $t = \frac{D(\theta^0)}{2}$, $\theta_1(t) = \theta_3(t)$. We have $\theta\left(\frac{D(\theta^0)}{2}\right) = \theta_2\left(\frac{D(\theta^0)}{2}\right) = \theta_3\left(\frac{D(\theta^0)}{2}\right)$

Then suppose the opposite, $\alpha < D(\theta^0)/2$. Since $\theta(t)$ is continuous, we have $\lim_{x \rightarrow \alpha} \theta(x) = \theta(\alpha)$. That is,

$$\theta(x) = \theta^0 + \begin{bmatrix} \omega + 1 \\ k(t) \\ \omega - 1 \end{bmatrix} x = \theta(\alpha), \text{ as } x \rightarrow \alpha$$

Since $\theta_1(\alpha) = \theta_3(\alpha)$, $\theta_1^0 + (\omega + 1)x = \theta_3^0 + (\omega - 1)x$. Solving for x, we get $x = (\theta_3^0 - \theta_1^0)/2 = D(\theta^0)/2$, which is a contradiction.

Thus, we get $\alpha = D(\theta)/2$ $\square$

5 N case

let $\Sigma = \{\theta | (\theta_1 - \theta_2) \dots (\theta_i - \theta_j) \dots (\theta_{N-1} - \theta_N) = 0\}$. Σ will split the space into some order, i.e. $D_1 = \{\theta_1 > \theta_2 > \dots > \theta_N\}$ whenever $\theta \notin \Sigma$. We claim that the solution will not cross Σ

Lemma 9. *Let $\theta(t)$ be any solution of the system with initial phase diameter satisfying $D(\theta^0) < \pi$. If $\theta^0 \in D_i$, then the solution will stay at $\overline{D_1}$, which will not cross Σ .*

Proof. WLOG, we suppose $\theta^0 \in D_1$ and at some time $t = \tau$ the solution touches Σ , i.e. $\theta(\tau) \in \Sigma$. let $H(\theta) = (\theta_1 - \theta_2) \dots (\theta_i - \theta_j) \dots (\theta_{N-1} - \theta_N)$. We will check $\frac{d}{dt} H(\theta(t))|_{t=\tau}$.

$$\frac{d}{dt} H(\theta(t)) = \nabla H(\theta) \cdot v \text{ where } v \in K[X](\theta)$$

We can see that $\frac{\partial}{\partial \theta_i} H(\theta) = \sum_{k=1}^{i-1} (-1) \frac{H(\theta)}{\theta_k - \theta_i} + \sum_{k=i+1}^N \frac{H(\theta)}{\theta_i - \theta_k} = \sum_{k=1, k \neq i}^N \frac{H(\theta)}{\theta_i - \theta_k}$
 Since $\theta(\tau) \in \Sigma$, suppose $\theta_i(\tau) = \theta_j(\tau)$, we get

$$\frac{\partial}{\partial \theta_i} H(\theta) = -\frac{\partial}{\partial \theta_j} H(\theta)$$

Note that if $\theta_i(\tau) = \theta_j(\tau) = \theta_p(\tau) \dots$, we have $\frac{\partial}{\partial \theta_t} H(\theta(\tau)) = 0, \forall t$. It is easy to see that $\frac{d}{dt} H(\theta(\tau)) = 0$ or $\emptyset$. Since $\theta_i(\tau) = \theta_j(\tau)$, we rewrite

$$X(\theta(\tau)) = \begin{bmatrix} \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_1 - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_i - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_j - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_N - \theta_k)) \end{bmatrix} = \begin{bmatrix} \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_1 - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_i - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_i - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_N - \theta_k)) \end{bmatrix}$$

for $v = (v_1, v_2, \dots, v_N) \in K[X](\theta(\tau))$, if $\sum_{k=1}^N \sin(\theta_h - \theta_k) = 0$, $v_h \in [\omega - 1, \omega + 1]$. Thus, $\frac{d}{dt} H(\theta) = \nabla H(\theta(\tau)) \cdot v = 0$ or $\emptyset$. That is, the solution will stay at Σ once it touches it.

Suppose at some time $\tilde{t}$ the solution leaves D_1 . Since θ is continuous, by IVT, we have $H(\theta(c)) = 0$, for $c \in [0, \tilde{t}]$ and from the previous we know it stays at Σ . $\square$

This shows that the evolution of the θ will not change its order. That is, for θ^0 with $\theta_1^0 \geq \theta_2^0 \geq \dots \geq \theta_N^0$ stays the same $\theta_1(t) > \theta_2(t) > \dots > \theta_N(t)$ for all t .

Then suppose $\theta^0 \in \overline{D_1}$. By lemma 9. we have θ_1 being largest and θ_N be the smallest for all time. That is, we can rewrite the system as

$$X(\theta) = \begin{bmatrix} \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_1 - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_i - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_j - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_N - \theta_k)) \end{bmatrix} = \begin{bmatrix} \omega - 1 \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_i - \theta_k)) \\ \vdots \\ \omega - \text{sgn}(\sum_{k=1}^N \sin(\theta_j - \theta_k)) \\ \vdots \\ \omega + 1 \end{bmatrix}$$

Therefore, we only need to check when $\theta_1(t) = \theta_N(t)$, from the order then we know $\theta_1(t) = \theta_2(t) = \dots = \theta_N(t)$.

$$\begin{aligned} \theta_1(t) &= \int_0^t \omega - 1 \, dt + \theta_1^0 = \int_0^t \omega + 1 \, dt + \theta_N^0 = \theta_N(t) \\ \implies (\omega - 1)t - (\omega + 1)t &= \theta_N^0 - \theta_1^0 \implies t = \frac{\theta_1^0 - \theta_N^0}{2} = \frac{D(\theta^0)}{2} \end{aligned}$$

which is the converge time T . Also,

$$\theta_i(t) = \frac{\theta_1 + \theta_N}{2} + \omega t, \quad \forall t > T$$

6 Finite time convergent gradient flows

Theorem 3 (Lasalle invariance principle). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be a locally Lipschitz and regular function. Let $x_0 \in S \subset \mathbb{R}^d$, with S compact and strongly invariant. Assume either $\max \tilde{\mathcal{L}}_X f(x) \leq 0$ or $\tilde{\mathcal{L}}_X f(x) = \emptyset$ for all $x \in S$. Let $Z_{X,f} = \{x \in \mathcal{R}^d | 0 \in \tilde{\mathcal{L}}f(x)\}$. Then, any solution $x : [t_0, +\infty) \rightarrow \mathbb{R}^d$ starting from x_0 converges to the largest weakly invariant set M contained in $\bar{Z}_{X,f} \cap S$. Moreover, if the set M is a finite collection of points, then the limit of all solutions starting at x_0 exists and equals one of them.*

Theorem 4 (Finite-time convergence with first-order information). *Under the same assumptions of Theorem 3, further assume that there exists a neighborhood U of $Z_{X,f} \cap S$ in S such that $\max \tilde{\mathcal{L}}_X f \leq -\epsilon < 0$ a.e. on $U \setminus (Z_{X,f} \cap S)$. Then, any solution $x : [x_0, +\infty) \rightarrow \mathbb{R}^d$ starting from S reaches $Z_{X,f} \cap S$ in finite time. Moreover, if $U=S$, then the convergence time is upper bounded by $(f(x_0) - \min_{x \in S} f(x))/\epsilon$*

Corollary 4.1. *If $\tilde{\mathcal{L}}_X f = -\epsilon < 0$ a.e. on $U \setminus (Z_{X,f} \cap S)$ and $U=S$, then the convergence time is $(f(x_0) - \min_{x \in S} f(x))/\epsilon$*

Consider $\dot{x} = \text{sign}(\nabla V(x)) = X(x)$. let $f(x) = \frac{1}{2}(\max_{i \in \{1,2,\dots,d\}} x_i - \min_{i \in \{1,2,\dots,d\}} x_i)$
When $\frac{\partial V}{\partial x_M}(x) < 0$ and $\frac{\partial V}{\partial x_m}(x) > 0$, the convergence time is $f(x_0)$